

Mashalakh Mamedov

Mechatronics Engineering MSc Student | BSc Electrical and Electronics Engineering | Power Electronics, MATLAB and Simulink Modelling and Validation

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+39 344 781 6935 | [linkedin.com/in/mashalakh](https://www.linkedin.com/in/mashalakh) | Available from October 2026, 6–12 months

MSc candidate in Mechatronics Engineering at the University of Rome Tor Vergata, with a BSc in Electrical and Electronics Engineering. I model and validate electromechanical systems in MATLAB, Simulink, Simscape Electrical and Autodesk Inventor — power conversion, robot dynamics, vehicle energy and image-based evaluation, in the same toolchain. Two years nine months of prior on-site industrial experience on a gas-pipeline expansion project. Seeking a 6–12 month engineering internship from October 2026 in power electronics or adjacent controls, model-based design, robotics or test and validation.

EDUCATION

MSc, Mechatronics Engineering — University of Rome Tor Vergata, Rome, Italy

Sep 2024 – expected 2027

Taught in English. Currently enrolled, second year.

- Coursework: Power Electronics and Electrical Drives, Robot Mechanics, Computer Vision, Integrated Sensors, VLSI Circuit and System Design, Fundamentals of Mechanics of Systems, Thermodynamics and Heat Transfer.

BSc, Electrical and Electronics Engineering — Tbilisi State University

Awarded Jul 2023

Ivane Javakhishvili Tbilisi State University, Tbilisi, Georgia. GPA 2.9 / 4.0. Senior Design Project: A (96).

SELECTED ENGINEERING PROJECTS

3 kVA Single-Phase H-Bridge Inverter

A.Y. 2025/26

MSc Power Electronics and Electrical Drives, Tor Vergata. Individual project. Simulation only — no prototype, PCB or laboratory measurement.

- Designed and simulated a 3 kVA single-phase H-bridge inverter, 400 V DC to 230 V RMS at 50 Hz with unipolar SPWM and a 5 kHz carrier, as a parameterised MATLAB and Simulink model in Simscape Electrical Specialized Power Systems.
- Sized a 5 mH / 20 μ F LC output filter and validated across unity, 0.5 inductive and 0.5 capacitive power factors plus no-load and full-load step transients, reaching **1.64% output voltage THD** at full load against a below-3% requirement.
- Diagnosed an LC filter resonance under inductive loading — the load current stayed smooth while the voltage rang — and cut THD from 3.64% to **2.33%** with a 5 Ω series damping resistor, reporting the resulting 0.72% to 1.64% rise at unity power factor as the cost of the fix.

Fire and Smoke Semantic Segmentation

Jun – Jul 2026

MSc Computer Vision, Tor Vergata. Two-person project; I owned the deep-learning workflow end to end.

- Built the dataset pipeline recovering binary masks from COCO polygon annotations, then trained DeepLabv3+ on a ResNet-18 backbone with transfer learning and inverse-frequency class weighting across a 141 / 40 / 20 split.
- Benchmarked the network against a teammate's classical colour-threshold and morphology baseline on identical test data and metric, reaching a **mean test F-measure of 0.8507 against 0.4421**.
- Traced a NaN mean F-measure to zero-true-positive images, which led to auditing the labels: one test image showing clear fire carried an empty reference mask, worth about **4.48 percentage points** of the test mean. Kept it in the results and reported it as a data-quality finding.

RoboLatte — UR5e Dynamic Simulation

Jan 2026

MSc Robot Mechanics, Tor Vergata. Two-person project; I owned the Autodesk Inventor workflow. Simulation and CAD only.

- Built the UR5e CAD assembly in Autodesk Inventor, drove all six revolute joints with imposed cubic ramp motions, and configured Dynamic Simulation with gravity, CAD-derived inertias and a 3 N end-effector payload.
- Extracted joint driving forces from the Output Grapher at three instants and unit-converted N·mm to N·m, deriving an offset joint-convention mapping so the results were comparable with my teammate's analytical MATLAB inverse dynamics.
- Five of six joints agreed in sign and order of magnitude; Joint 3 did not, traced to the approximate link inertias the MATLAB model was forced to use. Reported rather than smoothed over.

V2V-Inspired Eco-Driving Energy Analysis

2026

MSc Powertrain Technologies for Future Mobility, Tor Vergata. Two-person project; I owned the MATLAB and Simulink vehicle model. Simulation only.

- Built a longitudinal road-load and traction-energy model with a parallel Simulink block implementation, quantifying a **9.7% traction-energy saving** (881.5 to 795.6 kJ; 8.47 to 7.62 L/100 km) from an anticipating speed profile.
- Reconciled script and block model to 0.0% discrepancy after tracing the mismatch to central versus one-sided differentiation and aligning the solver — an implementation-consistency check, not independent validation.

Remotely Operated Mobile Manipulator

2023

BSc Senior Design Project, Tbilisi State University. Four-person team; individual task split not documented.

- Led and co-authored the technical report and contributed across Arduino-based control, system integration and troubleshooting for a commercial 4-axis arm integrated onto a DIY 4WD platform with an NRF24L01 2.4 GHz link, L298N drive electronics and a 3D-printed gripper.
- The team diagnosed motor-induced supply sag causing Arduino resets and resolved it by separating the motor and logic power domains.

EXPERIENCE

Mechanical Installation Support Technician & Interpreter — Bechtel-Enka JV <i>South Caucasus Pipeline Expansion (SCPX) — Gardabani, Georgia, on site.</i> — Worked from P&ID schematics and technical drawings on an active gas-pipeline expansion site, assisting with industrial machinery alignment during installation and shutdown work. — Interpreted on site between Siemens, Solar Turbines and Ingersoll Rand vendor engineers and Georgian crews, and tracked critical components through shutdown work under BP and Bechtel EHS standards.	Feb 2016 – Jul 2017
Warehouse Data Entry Clerk — Bechtel-Enka JV <i>Same SCPX project — Gardabani, Georgia. Two years nine months on site in total.</i>	Jul 2017 – Oct 2018
Data Annotation Specialist — ALLSTARSIT <i>Tbilisi, Georgia.</i> — Annotated autonomous-driving imagery in a proprietary Linux-based tool, covering 2D and 3D tasks for vehicles, pedestrians, traffic lights and road markings, to predefined annotation policies with QA and team-lead review workflows.	Jul 2024 – Oct 2024

TECHNICAL SKILLS

Modelling & simulation	MATLAB · Simulink · Simscape Electrical Specialized Power Systems · parameterised model design · solver selection · script-versus-model reconciliation
Power electronics	H-bridge inverter design · unipolar SPWM · LC filter design and damping · FFT/THD analysis · power-factor load modelling · semiconductor selection with loss and thermal estimation
Robotics & control	Forward and inverse kinematics · Denavit-Hartenberg modelling · workspace analysis · trajectory planning · inverse dynamics · CAD dynamic simulation
Computer vision	Semantic segmentation · DeepLabv3+ · ResNet-18 · transfer learning · class weighting · COCO-to-mask dataset preparation · F-measure evaluation · label-quality auditing
Automotive	Longitudinal vehicle dynamics · road-load and traction-energy modelling · model verification
Embedded & CAD	Arduino / C++ · NRF24L01 2.4 GHz · servo control · L298N motor drivers · Autodesk Inventor

LANGUAGES, CERTIFICATION AND ADDITIONAL EXPERIENCE

Languages	Turkish (native) · Azerbaijani (native) · English (C1 working proficiency) · Georgian (B2) · Russian (B1) · Italian (basic)
Certification	EF SET English Certificate — 74/100, C2 Proficient (Reading 75, Listening 72), awarded 30 January 2025
Additional	iGaming Customer Support Specialist, RISK Inc., Remote / Rome — May 2025 to present · Customer Service Specialist, Majorel, Tbilisi — Feb 2022 to Jun 2024